levied as fees against the merchant. Due to these extra costs, merchants may need to adjust prices to protect themselves from both legitimate and illegitimate disputed charges.

Cryptoasset transactions are irreversible; therefore chargebacks are impossible. While an irreversible transaction may sound scary, it actually benefits the efficiency of the overall system. With credit card chargebacks, everyone has to bear the cost, whereas with cryptoassets only those who are careless bear the cost.

Many claim that hacked exchanges are proof that cryptoassets are insecure, but this displays a fundamental misunderstanding of the software architecture. Recall the four layers of any blockchain ecosystem that we discussed in Chapter 2: decentralized hardware, cryptoasset software, applications, and users. It is the third layer, applications, that are targeted in the majority of hacks. Thus, an exchange, which is an application that runs on top of the cryptoasset software, gets hacked. The underlying blockchain performs its job perfectly and remains uncompromised. The same analogy can be applied to applications that run on Apple's operating systems. Just because one of the apps is hacked doesn't mean Apple's underlying operating system or hardware is insecure.

Understanding that it's the applications and exchanges that use and trade cryptoassets that are most susceptible to hacks, it's all the more important for the innovative investor to be